

Ref no:	SSOW7	Calders & Grandidge	
Compiled By: D. Losty	Safe System of Work Pole Fabrication	Rev: 6 Revision date: November 2023	
Effective Date: Feb 2014		Revised By: L. Reece	

REVISION RECORD

Issue	Date	Details
1	Feb 2014	Original document. Replaces SSOW12.
2	Feb 2014	Minor changes.
3	Apr 2014	Clause 6.1.2 amended to clarify PPE requirements for use of grinders.
4	June 2017	Changed Logo and JS Job Title and removed Ltd from Calders and Grandidge
5	May 2019	Changed Reviewed By and Authorised By. Restriction of 11.5m poles in Shed 4.
6	Nov 2023	Full Review

AUTHORISATION

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1. PURPOSE

- 1.1 To document the arrangements in place for the safe fabrication of BT and utility company poles within the fabrication sheds in addition to the safe loading and unloading of those sheds.

2. SCOPE

- 2.1 This Safe System of Work applies to loading, unloading and fabrication activities that take place within the fabrication sheds 1 to 4 and the 'Big Shed'.

3. GLOSSARY

Calders & Grandidge or (C&G)	Calders & Grandidge (Boston) Limited
Fabricator	Pole fabricator.
PPE	Personal Protective Equipment
COSHH	The Control of Substances Hazardous to Health Regulations
Cleansweep	Regular, scheduled program of cleaning and housekeeping activities, designed to deliver a safer working environment.

4. REFERENCES

RA027	Risk Assessment – Portable Power Tools
RA168	Risk Assessment – Use of Hammers
RA192	Risk Assessment – Use of Nail Guns
RA137	Risk Assessment – Fabrication of Poles & Associated Activities
RA140	Risk Assessment – Use of Chainsaw & Associated Activities

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5 RECORDS

Record No if applicable	Title	Access, Disposal Retention, Authorised Person	Location	Minimum Retention Period	Index Method
-	Daily Tool Check Sheet	Pole Yard Supervisor	Archive Room	5 Years	Date.
-	Clean Sweep Form	Pole Yard Supervisor	Archive Room	5 Years	Date & Zone.

6. PROCESS

6.1 General Requirements

6.1.1 Only fully trained and authorised personnel are permitted to operate Highland Bears, woodworking, and abrasive wheel machinery.

6.1.2 Appropriate PPE must be worn as indicated in the table below:

PPE Item	Chainsaw	Routers/ Planers/ Trenchers/ Drills	Air Gun	Angle/ Bench Grinder	Hammer
Hearing Protection	✓	✓	✓	✓	✓
Hard Hat	✓				
Protective Footwear	✓				
Chainsaw Gloves	✓				
Chainsaw Jacket	✓				
Chainsaw Trousers	✓				
Plastic Visor	✓ (with chin guard fitted)			✓*	
Mesh Forrester Visor #	✓ (with safety glasses under)				
Safety Glasses/ Face Shield		✓			✓
Safety Goggles			✓	✓*	
Protective Gloves		✓	✓	✓	✓
Chainsaw Gaiters	When cutting at ground level				
Dust Mask/ RPE	When working on treated wood	When working on treated wood			
Protective Coveralls	When working on treated wood	When working on treated wood		✓	

= When working on treated wood, plastic face visors must be used instead of mesh visors.

* = When using grinders, staff must either wear a plastic visor or safety goggles – whichever is most suited to the task.

6.1.3 All pole fabrication sheds are designated as hearing protection zones and, as such, hearing protection must be worn at all times in these areas.

6.1.4 Sun barrier cream should be applied where personnel are exposed to the sun or working on treated timber.

6.1.5 Storage lockers are provided for tooling and PPE and all Fabricators are required to safeguard their equipment.

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- 6.1.6 Shadow boards are in place for the organised placement and retrieval of key items of equipment. Fabricators must ensure that these boards are properly utilised and that used equipment is returned to its designated location throughout each working day.
- 6.1.7 Operators must complete daily pre-use inspections for all tools and equipment to be used, including PPE (and record this on the daily inspection sheet). Defects must be reported immediately to the supervisor or line manager and replacements obtained.
- 6.1.8 Woodworking operator instruction manuals and COSHH data sheets are located in the Pole Production office and are available for inspection. Operators must familiarize themselves with these documents.
- 6.1.9 Working areas must be kept clear at all times and good housekeeping practices followed (waste bins are provided). These arrangements must be monitored by the Supervisor/ Line Manager. (areas must be maintained in line with Cleansweep action/ audit sheets displayed in each fabrication area).
- 6.1.10 Only authorised personnel are permitted inside the fabrication sheds. (Management personnel and persons accompanied by management, all pole yard staff, first aiders and any persons given authorisation by pole yard Production Manager/Supervisor).
- 6.1.11 All hot work must cease at least 60 minutes before the end of the shift. During the final 60 minutes, the area in which the hot work occurred must be monitored to ensure that a fire does not occur.
- 6.1.12 Climbing onto poles/benches is not permitted. When navigating the sheds, personnel must step over benches in one smooth movement.
- 6.1.13 Fabrication involves manual handling (rolling poles and sliding the tips of poles) and individual Fabricators will have varying capabilities in this regard. When Manual handling of poles is required this should be carried out as follows:
- a
 - Fabricators must manual handle individual poles in accordance with their training and capabilities.
 - b
 - If Fabricators cannot **easily** manual handle a pole, they must request assistance and not attempt to further manipulate the pole until such assistance is provided and they can do so safely.
 - c
 - Fabricators must take particular care when 'breaking down' poles that have been stacked by the Highland Bear driver on the in-feed chain bed. During such 'breaking down', Fabricators must stand clear of the butt of the pole and in such a position that the pole being manipulated will roll safely past them when released.
- 6.1.14 All tools, fuels and lubricants must be locked away every night in the Job Boxes and COSHH cabinets provided to the Pole Fabricators. Electric cables and air-lines must be rolled and hung up on the hooks provided.
- 6.1.15 Fabricators must ensure that the perimeter safety chains are connected across the shed entrances at all times.
- 6.1.16 Consumption of food and drink is not permitted inside the fabrication sheds.
- 6.1.17 Hardwood blocks or pads which are used to help spread the load of the Highland Bear Crane legs MUST not be left over walkways at any time.

6.2 Loading Requirements

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- 6.2.1 Highland Bear operators and Pole Fabricators are given instructions from their Supervisor or Line Manager in the form of a production plan. This allows the Highland Bear driver to load the correct poles into the fabrication sheds and in the order required. The pole Fabricators use the production plan in the course of production.
- 6.2.2 The Highland Bear operator/dressing machine operator must inform the pole Fabricator (prior to loading) that loading is about to commence and must check that there is sufficient space to accommodate the poles.
- 6.2.3 If there is insufficient space, the Supervisor or Line Manager must be contacted, for further instructions.
- 6.2.4 The pole fabricator or Roller crane banksman must operate the winching controls, bringing the poles into the shed. If they are not present then the Highland Bear operator should ask for assistance or winch the poles in themselves.

6.2.5 Maximum Loading Heights and Lengths

- 6.2.5.1 Benches must not be overloaded. The maximum loading height for poles on feed benches is as specified in the table below:

Table A			
Pole Type	Pole Length	Pole Class	Maximum Height
Home-grown and imported	Any	Stout	1 Layer
	11m or less	Medium	2 Layers
	More than 11m	Medium	1 Layer
	Any	Light	3 Layers

- 6.2.5.2 Poles must be placed on pole benches with the poles parallel to each other. Highland Bear operators must avoid placing poles across one another diagonally.
- 6.2.5.3 Pole lengths must not exceed those specified in the following table:

Table B	
Shed Number	Maximum Permitted Length (metres)
1	13
2	12 (11.5)
3	11* (10.5)
4	12** (11.5)
Big Shed	24
Auto Dresser	15
New Auto Dresser	18

* Due to the restricted working space, poles over 10 metres in length must not be fabricated in the area between the out feed end of the middle chains and the out feed end of Shed 3.

** Due to restricted working space, finished product in Shed 4 must not exceed 11.5 metres in length

Sheds 2,3,4 may be required to go to the maximum length possible if there are any unforeseen circumstances restricting the preferred sheds - this will be discussed with the operatives prior to works commencing

- 6.2.6 Highland Bear operators must follow all instructions regarding the safe use of the crane, laid out in the ALLMI Operators Manual (available to all operators).

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- 6.2.7 Prior to offloading from the Highland Bear, the Highland Bear operator must ensure that no personnel are within the crane's extended circle, including outside the rear of the shed being loaded.
- 6.2.8 The Highland Bear operator must never lift any load over the head of any personnel.
- 6.2.9 When loading poles into Shed 4 with the Hyland Bear, the operator must ensure that all personnel are kept a safe distance away, as the poles will be lifted over the shed itself. During this operation:
- a
 - Physical barriers must be used to prevent access to the operational areas in which the crane is operating (such as safety barriers or safety chains).
 - b
 - These barriers or chains must only be crossed with the permission of the crane operator, who must first place the crane in a parked state (with no raised boom or load).
- 6.2.10 The Highland Bear operator must load the feed beds safely and securely to prevent poles from falling from the beds. The poles must not be overlapped or dropped onto the benches.

6.3 Chain Benches Safe Working

- 6.3.1 Pole Fabricators working in the sheds must isolate the power to the chain benches when they do not want them to be operated automatically by the Automatic dressing machine operator or any member of the roller crane team.
- 6.3.2 Pole Fabricators must never carry out any fabricating work on the in-feed chain benches unless they have isolated the bench in advance (*this is to ensure that the chains cannot be accidentally operated by the Dressing Machine Operator*).

6.4 Loading or Unloading Over a Walkway or Roadway

- 6.4.1 The Highland Bear operator must chain off (or place barriers/cones across) any areas where loading or unloading operations take place over any walkway or roadway, to prevent authorised access to the operational area.

6.5 Fabrication Process – Pre-fabrication Actions

- 6.5.1 At the start of the working day, during the daily Supervisor's Briefing, the supervisor must:
- a
 - Issue Pole Production Sheets to Fabricators
 - b
 - Brief Fabricators on the work requirements and sequence for the day, using production plan for detail
 - c
 - Brief staff on the details of any recent accidents, incidents and near misses (including environmental events)
 - d
 - Issue specific instructions for the day.
 - e
 - Invite questions from Fabricators and encourage participation in the process.
 - f
 - Check the condition of the PPE of everyone present at the briefing.
- 6.5.2 During the Supervisor's Briefing, Fabricators must:

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- a
 - Listen carefully to the information being given.
- b
 - Participate and seek clarification for anything that is unclear.
- 6.5.3 Following the briefing, Fabricators must proceed to their allocated shed and not commence any work until they have
 - a
 - Switched on lights and ensured sufficient illumination is available.
 - b
 - Conducted a visual inspection and functional test of all tools, cabling and devices that will be used - and recorded the results of these on the daily tool check sheet.
 - c
 - Conducted a visual inspection and functional test of all chain benches that will be used - and recorded the results of these in the designated defect book for each item.
 - d
 - Checked that the floor area is clear of slipping and tripping hazards.

6.6 Fabrication Process – Fabricating Poles

- 6.6.1 Following checks, the Fabricator must consult the pole production plan and any other written instructions and commence fabricating as indicated.
- 6.6.2 If poles are already present in the fabrication shed, and those poles are not listed on the plan or through other documentation that has been provided, then clarification must be sought from the Supervisor/line Manager.
- 6.6.3 Where two or more Fabricators are to work in the same shed simultaneously, each Fabricator must agree with the other(s) what function they will perform and a clear understanding must be reached with regard to roles.
- 6.6.4 At least 3m must be maintained between a chainsaw operator and any other person whilst the saw is in operation (unless the other person is wearing full chainsaw PPE)
- 6.6.5 Any person wishing to communicate with a Fabricator who is using a chainsaw, must wait at a safe distance until the Fabricator has stopped chainsawing, has applied the chainsaw brake and has indicated that it is safe to approach.
- 6.6.6 During the work, Fabricators must:
 - a
 - Correctly wear PPE as specified in the table at 6.1.2.
 - b
 - Keep the work area tidy and regularly remove waste to reduce underfoot hazards.
 - c
 - Practice good tool management to minimise risks from trailing cables.
 - d
 - Monitor tool performance during use and replace worn bits, blades etc. as necessary to ensure optimum performance. Clean tools immediately after working on treated poles
 - e
 - Remain properly hydrated and take measures to cool/ warm themselves as necessary to perform their duties safely during extremes of weather.
- 6.6.7 Fabricators must not use any tool or undertake any work process for which they have not been trained.

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6.6.8 Fabrication must be conducted in a logical sequential manner that minimises possible confusion between Fabricators.

6.6.9

- a Fabricators operating the nail gun must take particular care with regard to personnel who may be around them during operation. Care must be taken to keep fingers away from the firing point
- b In some instances, a hookaroon/cant hook will be required generally for bigger diameter poles to aid with rolling and positioning of a pole on the chain benches, the fabricators must be shown initially by an experienced operator on how to use the tool correctly, the handle should never be directly in line with the users face

6.7	Fabrication Process – Post-fabrication Actions
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6.7.1 On completion of Fabrication activities, the Fabricator must:

- a
 - Collect up and safely stow all power tools and transformers in the secure tool storage container provided, using chainsaw covers to protect blades and cabling from damage.
- b
 - Lock the secure tool storage container.
- c
 - Ensure that all tooling allocated to the shadow board is replaced in its designated location.
- d
 - Conduct Cleansweep to ensure that the shed is left in a safe and tidy condition for the next day.
- e
 - Ensure that all flammable liquids and substances are securely stored.
- f
 - Switch off all appliances and lights when leaving the shed (unless specifically instructed otherwise).